

Module: `Problem 3 - Printing Out all Available Letters`

This module provides a utility for the classic Hang-man style games. It computes which letters of the English alphabet have **not** been guessed yet, based on a list of previously guessed characters. The function operates only on lowercase letters `'a' - 'z'` and returns the remaining letters as a concatenated string.

Functions

`getAvailableLetters(lettersGuessed)`

Description

Returns a string containing all lowercase English letters that are **not** present in the supplied `lettersGuessed` list. The function iterates through the alphabet in order and appends each letter that has not been guessed.

Parameters

- `lettersGuessed` (*list of str*): A collection of characters (or strings) that have been guessed so far. The function expects each element to be a single lowercase letter, but it will treat any non-matching entry as “not guessed”.

Returns

- *str*: A string of the remaining (available) letters, ordered from `'a'` to `'z'`.

Examples

```
# Example 1 - no letters guessed yet
result = getAvailableLetters([])           # → 'abcdefghijklmnopqrstuvwxyz'

# Example 2 - some letters guessed
result = getAvailableLetters(['a', 'b', 'c', 'd', 'e']) # → 'fghijklmnopqrstuvwxyz'

# Example 3 - input contains whole words (treated as non-letters)
result = getAvailableLetters(['apple', 'banana', 'cherry']) # → 'abcdefghijklmnopqrstuvwxyz'

# Example 4 - single-character list inside another list (as observed in tests)
result = getAvailableLetters(['a', 'b', 'c'][0]) # → 'defghijklmnopqrstuvwxyz'

# Example 5 - mixed valid and invalid entries
result = getAvailableLetters(['t', 'e', 's', 't', 'word']) # → 'abdcdfghijklmnopqruvwxyz'
```

Edge Cases / Notes - Non-letter entries: Elements that are not single lowercase letters (e.g., full words like `'hello'`) are ignored, so the full alphabet is returned. - **Case sensitivity:** The function only checks against lowercase `'a' - 'z'`. Uppercase letters will be considered “not guessed”. - **Duplicate guesses:** Re-guessing the same letter has no effect; the letter remains excluded from the result. - **Empty input:** An empty `lettersGuessed` list yields the complete alphabet. - **Performance:** The implementation runs in $O(26 + n)$ time, where n is the length of `lettersGuessed`, which is negligible for typical game use.